Candidate seat No: _______________

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B. Tech. Theory Examination (IT)

Apr-May 2018

MA244 Statistical and Numerical Techniques

Date: 09.05.2018, Wednesday   Time:10: 00 a.m. to 01: 00 p.m.   Maximum Marks: 70

Instructions:

i. The question paper contains two sections.
ii. Section I and II must be attempted in separate answer sheets.
iii. Figures to the right indicate marks.
iv. Make suitable assumptions and draw neat figure where it is required.
v. All notations and terminologies are standard.

Section-I

Q-1 Select the correct option for following multiple choice questions. [07]

1) The standard deviation of n values of variable is 17. What will be the new standard deviation if all values of variable increased by 10? **01**

a) 22 **b)** 12 **c)** 17 **d)** 10

2) Which of the following is true if A is an independent event from the event B? **01**

a) B may or may not be an independent event from the event A.

b) $P(A|B) = P(A) \& P(B|A) \neq P(B)$.

c) $P(A|B) \neq P(A) \& P(B|A) = P(B)$.

d) $P(A|B) = P(A) \& P(B|A) = P(B)$.

3) If $F(x)$ represent the cumulative probability function for the random variable X then $P(a \leq X \leq b) =$ _______. **01**

a) $F(a)+F(b)$ **b)** $F(b)-F(a)$ **c)** $F(a)-F(b)$ **d)** $F(b-a)$

4) For a Uniform random variable defined on $[a,b]$, the mean is given by ______. **01**

a) $\frac{a+b}{2}$ **b)** $\frac{b-a}{2}$ **c)** $\frac{a-b}{2}$ **d)** $\frac{(b-a)^2}{12}$

5) Any function of the random variable consisting a random sample is called a _____. **01**

a) Sampling distribution **b)** Statistic

c) Sample mean **d)** Sample variance

6) If r denote the Karl Pearson coefficient of correlation for linear relationship between two variables then which of the following is not true. **01**

a) $r^2 \to 0$ implies little or no linear relationship between two variables.

b) $r^2 > 0.8$ implies strong linear relationship between two variables.

c) $r^2 = 0$ implies lack of association between two variables.

d) $r^2 < 0.5$ implies weak linear relationship between two variables.

7) Which of the following is true for Spearman's coefficient of rank correlation (ρ) for given two rankings? **01**

a) $-1 \le \rho \le 1$.

b) Sum of square of differences of two rankings must be equal to zero.

c) If $\rho = 1$ then one ranking is reverse of the another ranking.

d) If $\rho = -1$ then both rankings are identical.

Q-2 Attempt any four out of six. **[16]**

1) Evaluate harmonic mean and standard deviation for the thickness (in mm) of 7 different materials manufactured continuously before being cut and wound into large rolls: **04**

32.2 30.4 31.0 31.2 30.3 30.7 31.2

2) The following scores represent the final examination grades of 60 students for an elementary statistics course: **04**

23	60	79	32	57	74	52	70	82
36	80	77	81	95	41	65	92	85
55	76	52	10	64	75	78	25	80
98	81	67	41	71	83	54	64	72
88	62	74	43	60	78	89	76	84
48	84	90	15	79	34	67	17	82
09	74	63	80	85	61			

Construct the frequency distribution table for the given data having 7 disjoint continuous classes of width 15 with right endpoint convention starting from 0. Also calculate relative frequency for each class and median for the distributed data.

3) Define conditional probability. If the probability that a regularly scheduled flight departs on time is 0.83, the probability that it arrives on time is 0.82 and the probability that it departs and arrives on time is 0.78 then find the probability that a plane **04**

i. arrives on time given that it departed on time.

ii. departed on time given that it has arrived on time.

4) A lab network consisting of 8 computers was attacked by a computer virus. This virus enters each computer with probability 0.7 independently of other computer. Compute the probability that virus enters in **04**

i. Exactly three computers.

ii. Atmost two computers.

5) Define the Poisson random variable . The number of customers arriving per hour at a certain automobile service facility is assumed to follow a Poisson distribution with $\lambda = 2$. Compute the probability that more than 3 customers will arrive in a 2 - hour period. **04**

6) State Central Limit Theorem and use it to approximate the probability that the sample mean of the weights of a chosen sample of 36 workers lies between 63 and 70. If the weights of a population of workers have mean 67 and standard deviation 9. **04**

Q-3

1) **Attempt any two out of three.** **[06]**

a) A laboratory blood test is 99 percent effective in detecting a certain disease when it is, in fact, present. However, the test also yields a “false positive” result for 1 percent of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply he or she has the disease.) If 0.5 percent of the population actually has the disease, what is the probability a person has the disease given that his test result is positive? **03**

b) Suppose that the number of cars X that pass through a car wash on any sunny Friday has the following probability distribution: **03**

x	5	6	7	8	9
$P(X=x)$	$\frac{1}{12}$	$\frac{1}{3}$	$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{6}$

Evaluate expected value and variance for X.

c) Consider n number of observations for two pair of variables (x,y) and (u,v) where $u_i = ax_i + b$ and $v_i = cy_i + d$ If r_{xy} & r_{uv} denote Karl Pearson's correlation coefficient for (x,y) and (u,v) respectively then prove $r_{uv} = \frac{a \cdot c}{|a| \cdot |c|} r_{xy}$. **03**

2) Eight students were ranked for their reading and writing ability of French language as follow: **[06]**

Student	A	B	C	D	E	F	G	H
Reading	3	4	7	2	5	8	6	1
Writing	6	1	3	5	7	2	8	4

Measure the degree of correspondence between two rankings by Kendall's coefficient of rank correlation (τ) and Spearman coefficient of rank correlation (ρ).

OR

2) A study was done to study the effect of ambient temperature X on the electric power consumed by the plant Y. Other factors were held constant, and the following data were collected from an experimental plant: **[06]**

$X(^\circ F)$	26	45	72	58	31	60	34	74
$Y(BTU)$	250	285	320	295	265	298	267	324

Find the equation of the regression line to predict the electric power consumed by the plant for an ambient temperature.

Section-II

Q-4 Select the correct option for following multiple choice questions. **[07]**

1) The number obtained by rounding off the number 21.475345 to four significant figure is ______. **01**

a) 21.475 **b)** 21.48 **c)** 21.47 **d)** 21.46

2) Newton's backward interpolation formula is used for ______ intervals. **01**

a) Open **b)** Unequal **c)** Equal **d)** Closed

3) The operator E is equal to __________. **01**

a) $1+\Delta$ **b)** $\nabla+\Delta$ **c)** $\nabla\Delta$ **d)** $\nabla-\Delta$

4) One of the normal equation for fitting of a straight line $y=a+bx$ is ______. **01**

a) $\sum y_i = na + b\sum x_i$ **b)** $\sum y_i = n^2 a + b\sum x_i^2$

c) $\sum y_i = na + b\sum x_i^2$ **d)** $\sum y_i = a + b\sum x_i$

5) In which method we approximate the curve of solution by the tangent in each interval? **01**

a) Taylor's method **b)** Euler's method

c) Runge - Kutta method **d)** None of these

6) Rate of convergence of Newton Raphson method is _______. **01**

a) 1 **b)** 1.618 **c)** 2 **d)** 0

7) In Simpson's 1/3 rule whole range is divided into ______ intervals. **01**

a) Odd **b)** Even **c)** Multiple of 3 **d)** None of these

Q-5 Attempt any four out of six. **[16]**

1) Find a second degree polynomial which passes through the points (1, - 1), (2, - 2), (3, - 1) and (4, 2) using proper interpolation formula. Find y, when x = 1.75 and 5. **04**

2) Using Lagrange interpolation formula, find the value of y corresponding to x = 10 from the following given data: **04**

x	5	6	9	11
$y = f(x)$	380	-2	196	508

3) An experimental data about force and velocity for an object suspended in a wind tunnel is given as: **04**

Velocity(m/s)	10	20	30	40	50	60	70	80
Force (N)	24	68	378	552	608	1218	831	1452

Use least-squares regression to determine the coefficients a and b in the function $y=a+bx$ that best fits the data. Also estimate the force when velocity is 55m/s.

4) Using Newton's divided difference formula compute *f(3)* from **04**

x	0	1	2	4	5	6
$f(x)$	1	14	15	5	6	19

5) Use Runge - Kutta method of order four with $h = 0.1$ to find the approximate solution of $\frac{dy}{dx} = x^2 + y; y(0) = -1$ at $x = 0.1 \& 0.2$. **04**

6) Evaluate the integral $\int_0^{1.2} e^x dx$ taking $n = 6$ using Trapezoidal and Simpson's 1/3 rule. **04**

Q-6

1) **Attempt any two out of three.** **[06]**

a) Derive a Newton Raphson iteration formula for finding the cube root of positive number N and use it to approximate the value of $\sqrt[3]{12}$. **03**

b) Use Taylor's series method for the equation $\frac{dy}{dx} = 3x + y^2$ to approximate y when $x = 0.1,$ given that $y = 1$ when $x = 0$. **03**

c) Solve the following system of equations by Gauss Jordan method : **03**

$$x + 3y + 2z = 17, \quad x + 2y + 3z = 16, \quad 2x - y + 4z = 13$$

2) Determine the equation to the best fitting exponential curve of the form $y = ae^{bx}$ for the data: **[06]**

x	1	3	5	7	9
y	115	105	95	85	80

OR

2) Perform five iterations of Gauss Seidel method to solve the system of equations: **[06]**

$$8x + 2y - 2z = 8, \quad x - 8y + 3z = -4, \quad 2x + y + 9z = 12$$